

Direct reciprocity

1. A reactive strategy (p, q) in the Prisoner's Dilemma means ...
 - (a) cooperate with probability p after the opponent cooperated, and q after they defected
 - (b) always cooperate with probability $p + q$
 - (c) play p against cooperators and q against everyone else
 - (d) cooperate on the first p rounds, then defect
2. Tit For Tat is the reactive strategy ...
 - (a) $(0, 0)$
 - (b) $(1/2, 1/2)$
 - (c) $(1, 0)$
 - (d) $(1, 1)$
3. When two reactive strategies play, the pair of actions each round forms ...
 - (a) a Markov chain on the states (CC, CD, DC, DD)
 - (b) a zero sum game
 - (c) a single fixed outcome
 - (d) an extensive form game
4. The stationary distribution of this Markov chain gives ...
 - (a) the Nash equilibrium of the stage game
 - (b) each player's first move
 - (c) the long-run fraction of rounds spent in each state, used to find average payoffs
 - (d) the number of rounds the game lasts
5. Why does Tit For Tat need care with the stationary-distribution method?
 - (a) its values $(1, 0)$ lie on the boundary, so the chain may not be ergodic
 - (b) it is not a reactive strategy
 - (c) it always cooperates
 - (d) it has more than two actions
6. A Markov chain is ergodic if it is ...
 - (a) zero sum
 - (b) irreducible and aperiodic, so it has a unique stationary distribution
 - (c) finite
 - (d) symmetric

7. A reactive strategy conditions its move on ...
 - (a) the opponent's action in the previous round only
 - (b) its own first move
 - (c) the round number
 - (d) the entire history of play
8. In the state labelling (CC, CD, DC, DD) , the state CD means ...
 - (a) player 1 defected and player 2 cooperated
 - (b) both players cooperated
 - (c) both players defected
 - (d) player 1 cooperated and player 2 defected
9. A player's long-run average payoff is computed as ...
 - (a) the payoff in the first round
 - (b) the number of cooperative rounds
 - (c) the stationary distribution weighted by that player's stage payoffs
 - (d) the largest stage payoff
10. Reactive strategies are a subclass of ...
 - (a) zero-determinant strategies only
 - (b) pure strategies only
 - (c) memory-one strategies
 - (d) dominant strategies
11. Generous Tit For Tat is the reactive strategy ...
 - (a) $(0, 1)$
 - (b) $(1, 1)$
 - (c) $(0, 0)$
 - (d) $(1, \epsilon)$ for a small $\epsilon > 0$
12. The stationary distribution π of a transition matrix M satisfies ...
 - (a) $\pi M = \pi$
 - (b) $\pi = M$
 - (c) $\pi M = 0$
 - (d) $M\pi = M$
13. When Tit For Tat meets Always Defect (both starting by cooperating), play settles at ...

- (a) mutual defection
 - (b) mutual cooperation
 - (c) alternating cooperation and defection forever
 - (d) Tit For Tat always winning
14. The first move of a reactive strategy is ...
- (a) always defection
 - (b) irrelevant to the long run
 - (c) set separately, since there is no previous round to react to
 - (d) determined by q
15. Tit For Tat does well in tournaments because it is ...
- (a) purely random
 - (b) never willing to cooperate
 - (c) always the first to defect
 - (d) nice, retaliatory and forgiving
16. In a reactive strategy (p, q) , the value p is the probability of cooperating when the opponent ...
- (a) has never moved
 - (b) defected on the previous round
 - (c) cooperates on the current round
 - (d) cooperated on the previous round
17. Always Defect is the reactive strategy ...
- (a) $(0, 0)$
 - (b) $(1, 1)$
 - (c) $(1, 0)$
 - (d) $(1/2, 1/2)$
18. The Markov chain of two reactive strategies has state space ...
- (a) $\{0, 1\}$
 - (b) $\{C, D\}$
 - (c) $\{CC, CD, DC, DD\}$
 - (d) $\{CC, DD\}$
19. In the closed-form stationary distribution, the quantity r_1 is defined as ...
- (a) pq

- (b) p/q
- (c) $p - q$
- (d) $p + q$

20. Zero-determinant strategies are memory-one strategies that ...

- (a) are not reactive (they condition on the full previous outcome)
- (b) ignore all history
- (c) are the same as Tit For Tat
- (d) cooperate with probability 1